Table 1: Correction payloads, paths, and tags used in the analysis.

Era	Correction	Payload / path	Key / tag
Run3_2022	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer22_22Sep2023_V4_MC; JEC data: Summer22_22Sep2023_V4_DATA; JER: Summer22_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22CDSep23-Summer22-NanoAODv12/latest/puWeights.json.gz	Collisions2022_355100_357900_eraBCD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22CDSep23-Summer22-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jetvetomaps.json.gz	Summer22_23Sep2023_RunCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22CDSep23-Summer22-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2022EE	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer22EE_22Sep2023_V4_MC; JEC data: Summer22EE_22Sep2023_Run{}_V4_DATA, Summer22EE_22Sep2023_V4_DATA; JER: Summer22EE_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/puWeights.json.gz	Collisions2022_359022_362760_eraEFG_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jetvetomaps.json.gz	Summer22EE_23Sep2023_RunEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2023	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer23Prompt23_V4_MC; JEC data: Summer23Prompt23_Run{}_V4_DATA, Summer23Prompt23_V4_DATA; JER: Summer23Prompt23_RunCv1234_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23CSep23-Summer23-NanoAODv12/latest/puWeights.json.gz	Collisions2023_366403_369802_eraBC_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23CSep23-Summer23-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jetvetomaps.json.gz	Summer23Prompt23_RunC_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23CSep23-Summer23-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2023BPix	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer23BPixPrompt23_V4_MC; JEC data: Summer23BPixPrompt23_Run{}_V4_DATA, Summer23BPixPrompt23_V4_DATA; JER: Summer23BPixPrompt23_RunD_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/puWeights.json.gz	Collisions2023_369803_370790_eraD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload

Continued on next page

Table 1 – continued

Era	Correction	Payload / path	Key / tag
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jetvetomaps.json.gz	Summer23BPixPrompt23_RunD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2024	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt24_V5_MC; JEC data: Summer24Prompt24_V5_DATA; JER: Summer24Prompt24_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/puWeights_BCDEFGHI.json.gz	Collisions24_BCDEFGHI_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing_V
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt24_RunBCDEFGHI_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2024_378981_386951_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2025	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt25_V3_MC; JEC data: Summer24Prompt25_V3_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/afs/cern.ch/work/v/vdamante/Hmm_newSkim/corrections/data/MUO/SF/Run3-25Prompt-Summer24-NanoAODv15/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt25_RunCDEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-25Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2025_391658_398903_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2026	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt26_V2_MC; JEC data: Summer24Prompt26_Run{}_V1_DATA, Summer24Prompt26_V1_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-26Prompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt26_RunBCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-26Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2026_401624_403937_golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium

Table 2: Selections and corrections applied during NanoAOD skimming, in execution order.

<table>
<tr><th>Step</th><th>Operation</th><th>Kind</th><th>Condition / definition</th><th>Scope</th><th>Correction state</th></tr>
<tr><td>1</td><td>Orthogonal luminosity split</td><td>Event filter</td><td>OrthogonalEraTag == 0/1/2 (seed 12345; luminosity fractions 110/111/26)</td><td>MC 2024–2026</td><td>Before all physics corrections</td></tr>
<tr><td>2</td><td>Generator VBF phase space</td><td>Definition only</td><td>Compute GenVBFFilter for selected inclusive/VBF-filtered DY samples; current code does not call df.Filter</td><td>MC DY 2024–2026</td><td>Before corrections; no rejection</td></tr>
<tr><td>3</td><td>Base event weights</td><td>Weight definition</td><td>Luminosity, generator sign, pileup and cross-section weights; configured QCD-scale sums</td><td>MC</td><td>Pileup is applied here; no rejection</td></tr>
<tr><td>4</td><td>Golden JSON</td><td>Event filter</td><td>Accept certified run:luminosityBlock pairs from maincfg.yaml</td><td>Data</td><td>Before object corrections</td></tr>
<tr><td>5</td><td>Muon ScaRe</td><td>Object correction</td><td>Momentum scale for data/MC; MC resolution smearing; NanoAOD and beam-spot-constrained momenta</td><td>Data and MC</td><td>Before every muon selection</td></tr>
<tr><td>6</td><td>Muon FSR recovery</td><td>Object correction</td><td>FSR photon recovery implemented in corrections/muon_fsr.py</td><td>Data and MC</td><td>After ScaRe; nominal selection uses Muon_pt_FSR_corr</td></tr>
<tr><td>7</td><td>JEC/JER</td><td>Object correction</td><td>Reapply JEC; apply JER to MC; produce JER/JES Total shifts when variations are enabled</td><td>Data and MC</td><td>Before MET and jet selections</td></tr>
<tr><td>8</td><td>MET quality flags</td><td>Event filter</td><td>Logical AND of goodVertices, globalSuperTightHalo2016Filter, EcalDeadCellTriggerPrimitiveFilter, BadPFMuonFilter, BadPFMuonDzFilter, hfNoisyHitsFilter, eeBadScFilter, ecalBadCalibFilter</td><td>Data and MC</td><td>NanoAOD flags; special 2022/2023 data bad-MET redefinition</td></tr>
<tr><td>9</td><td>Single-muon trigger matching</td><td>Event filter</td><td>HLT_IsoMu24; corrected+FSR muon pT > 26 GeV; trigger-object ID 13, filter bit 8, DeltaR < 0.4</td><td>Data and MC</td><td>After muon corrections; OR over nominal/shifts</td></tr>
<tr><td>10</td><td>Good muons</td><td>Object selection</td><td>corrected+FSR pT > 15.0 GeV; |eta| < 2.4; medium ID; pfIsoId >= 2</td><td>Data and MC</td><td>ScaRe+FSR; nominal and muon shifts</td></tr>
<tr><td>11</td><td>Exactly two muons</td><td>Event filter</td><td>Exactly two good muons; nominal and every muon shift must pass when variations are enabled</td><td>Data and MC</td><td>After ScaRe, FSR and trigger matching</td></tr>
<tr><td>12</td><td>Dimuon mass window</td><td>Event filter</td><td>50 < m_mumu < 200 GeV; nominal and every muon shift must pass</td><td>Data and MC</td><td>Selected ScaRe+FSR muons</td></tr>
<tr><td>13</td><td>Electron veto</td><td>Event filter</td><td>No electron with pT > 20 GeV, |eta| < 2.5 and Electron_mvaIso_WP90</td><td>Data and MC</td><td>Raw NanoAOD electron kinematics</td></tr>
<tr><td>14</td><td>Muon ID/ISO/trigger SF</td><td>Weight definition</td><td>Configured correctionlib SF sources for both selected muons; central and optional up/down</td><td>MC</td><td>After muon/electron filters; no rejection</td></tr>
<tr><td>15</td><td>Jet ID and b-tag flags</td><td>Object definition</td><td>Tight jet ID; b-tag flags at |eta| < 2.5; b-tag event-veto flag is stored but not filtered</td><td>Data and MC</td><td>JEC/JER jets and era-specific WPs</td></tr>
<tr><td>16</td><td>Jet veto map</td><td>Object definition</td><td>Evaluate map; apply_filter=False in analysis/skim.py</td><td>Data and MC</td><td>Corrected jets; no direct event filter</td></tr>
<tr><td>17</td><td>Selected jets</td><td>Object selection</td><td>corrected pT > 20.0 GeV; |eta| < 4.7; tight ID; outside veto map; DeltaR(j,mu1/2) > 0.4</td><td>Data and MC</td><td>JEC/JER jets; repeated for JER/JES shifts</td></tr>
<tr><td>18</td><td>VBF candidates</td><td>Definition only</td><td>Build pair outside the horn; store HasVBF and indices</td><td>Data and MC</td><td>No VBF event filter in skim</td></tr>
<tr><td>19</td><td>Snapshot</td><td>Output</td><td>Write surviving events and nominal/shifted columns</td><td>Data and MC</td><td>Final step; categories are downstream</td></tr>
</table>

Table 3: Era-dependent skim settings read from the YAML configuration.

<table>
<tr><th>Era</th><th>Muon/jet settings</th><th>Horn veto expression</th><th>Other event-filter settings</th></tr>
<tr><td>Run3_2022</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>(abs(v_ops::eta(Jet_p4)) > 2.5 && v_ops::pt(Jet_p4) < 50.0)</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24</td></tr>
<tr><td>Run3_2022EE</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24; special bad-MET runs=362433–362760</td></tr>
<tr><td>Run3_2023</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24; special bad-MET runs=366727–367144</td></tr>
<tr><td>Run3_2023BPix</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24</td></tr>
<tr><td>Run3_2024</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>((abs(v_ops::eta(Jet_p4)) >= 2.5 && abs(v_ops::eta(Jet_p4)) < 3.0) && v_ops::pt(Jet_p4) < 50.0)</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24</td></tr>
<tr><td>Run3_2025</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>(abs(v_ops::eta(Jet_p4)) < 0)</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24</td></tr>
<tr><td>Run3_2026</td><td>mu pT>15.0 GeV, jet pT>20.0 GeV; |eta_mu|<2.4, |eta_j|<4.7</td><td>abs(v_ops::eta(Jet_p4)) < 0</td><td>trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24</td></tr>
</table>

Important scope note. The `muons_selection`, `categories`, and `masses_regions` expressions in `config/Run3_*/selections.yaml` are not invoked by the current `analysis/skim.py`. They are downstream histogram-level definitions, not NanoAOD skim event filters.